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IP-Based Building Automation and Integration of BACnet, DALI, and KNX Systems

Design of an Integrated Smart Control System for a University Auditorium

1 Introduction

Smart Building Systems have witnessed significant development in recent years. Control of lighting and HVAC systems is no longer implemented as isolated subsystems, but rather as part of an integrated architecture aiming to improve comfort, enhance energy efficiency, and simplify the management of large-scale buildings.
In large educational facilities, especially large university auditoriums, there is a clear need for a control system characterized by:

- High reliability
- Fast response
- Local and centralized control capability
- Expandability and integration with different systems

This project focuses on designing an integrated control system for a large university auditorium using the following standard protocols:

- KNX as the central control core
- DALI as a dedicated lighting execution system
- BACnet as a professional HVAC management system
- IP Application as a supervisory layer

2 Definition of the Used Protocols and Their Differences

2.1 KNX Protocol

KNX is an international standard protocol for building automation based on the concept of distributed intelligence, where field devices are capable of making decisions without relying on a single central controller.
KNX is characterized by:

- Using Individual Address (IA) to identify each device

- Using Group Address (GA) to perform functions
- Supporting a wide range of devices such as:
  - Touch Panels
  - Push Buttons
  - Presence Sensors
  - Shutter Actuators
- Operating over multiple media such as KNX TP and KNX/IP

In this project, KNX is used as the main control system where all logical decisions are executed.

2.2 DALI Protocol

DALI (Digital Addressable Lighting Interface) is a digital protocol dedicated exclusively to lighting control.
Its main features include:

- Precise control of light intensity (dimming)
- Support for groups and scenes
- Diagnostic feedback from lighting drivers
- A single master per DALI line

DALI relies on:

- Short Address (0–63) for driver identification
- Groups (0–15) for collective control

In this project, DALI is used as the lighting execution layer, while the logical decision-making remains within KNX.

2.3 BACnet Protocol

BACnet is a standard protocol used in Building Management Systems and is widely adopted in HVAC applications.
BACnet relies on:

- Device Instance for device identification

- Objects such as:
  - Analog Value (AV)
  - Binary Output (BO)
- Services such as:
  - ReadProperty
  - WriteProperty
  - SubscribeCOV

In this project, the HVAC system operates entirely as a BACnet-native system.

2.4 IP Application Layer

The IP Application layer represents the highest level of the system and is used for:

- Monitoring
- Visualization
- Supervisory control and analytics

This layer does not communicate directly with execution protocols but interfaces exclusively with KNX through a KNX/IP Gateway.

3 Project Concept

The objective of this project is to design a smart control system for a large university auditorium such that:

- KNX acts as the central decision-making core
- Lighting execution is performed through the DALI lighting system
- HVAC execution is performed through the BACnet HVAC system
- Monitoring and management are handled via an IP Application

The system is based on:

- Distributed presence sensors across auditorium levels
- Manual control through switches and panels
- Block logic and manual override
- Intelligent operation based on actual occupancy

4 System Components

4.1 KNX Devices

Device Name	Function	Individual Address
KNX Touch Panel	Central Control Interface	1.2.1
KNX Presence Sensor – Level 1	Motion Detection	1.2.2
KNX Presence Sensor – Level 2	Motion Detection	1.2.3
KNX Presence Sensor – Level 3	Motion Detection	1.2.4
KNX Central Motion Sensor	Global Presence Detection	1.2.5
KNX 4-Rocker Switch	Manual Lighting Control	1.2.6
KNX 4-Rocker Switch	Manual Shutter Control	1.2.7
KNX Shutter Actuator	Shutter Control	1.2.8
KNX–DALI Gateway	Lighting Gateway	1.2.9
KNX–BACnet Gateway	HVAC Gateway	1.2.10
KNX/IP Interface	IP Connectivity	1.2.11
Thermostat	Manual HVAC Control	1.2.12

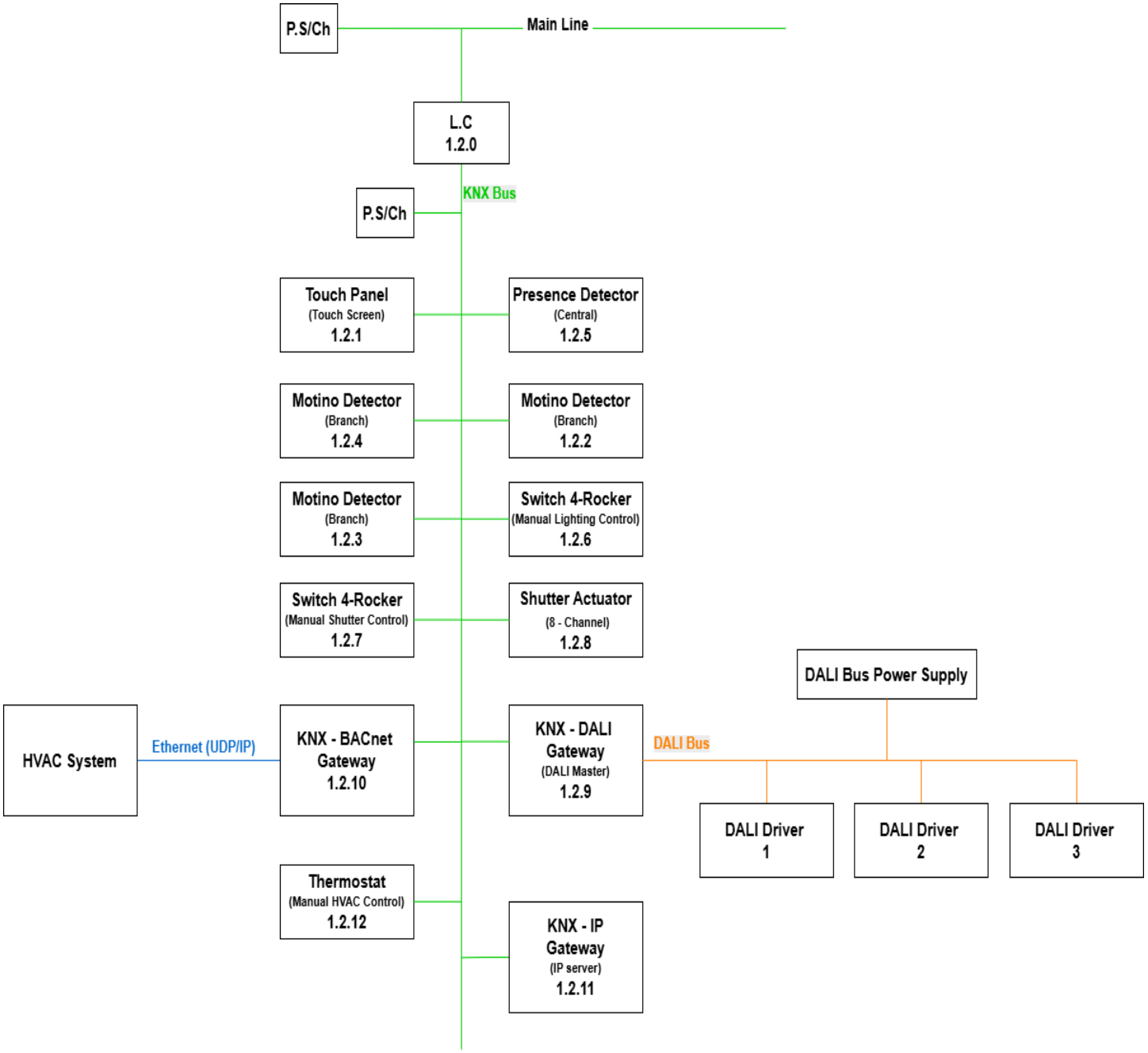
4.2 DALI Devices

Device	Function	Addressing
DALI LED Driver	Lighting Control	Short Address (0–63)

Drivers are distributed according to lighting lines and assigned to DALI groups.

4.3 BACnet Devices

Device	Function	Device Instance
BACnet HVAC Controller	HVAC Control	310001


P.S/Ch
Main Line
L.C
1.2.0
KNX Bus
P.S/Ch
Touch Panel
(Touch Screen)
1.2.1
Presence Detector
(Central)
1.2.5
Motino Detector
(Branch)
1.2.4
Motino Detector
(Branch)
1.2.2
Motino Detector
(Branch)
1.2.3
Switch 4-Rocker
(Manual Lighting Control)
1.2.6
Switch 4-Rocker
(Manual Shutter Control)
1.2.7
Shutter Actuator
(8 - Channel)
1.2.8
DALI Bus Power Supply
HVAC System
Ethernet (UDP/IP)
KNX - BACnet
Gateway
1.2.10
KNX - DALI
Gateway
(DALI Master)
1.2.9
DALI Bus
DALI Driver
1
DALI Driver
2
DALI Driver
3
Thermostat
(Manual HVAC Control)
1.2.12
KNX - IP
Gateway
(IP server)
1.2.11

5 KNX Group Addresses

5.1 Lighting

Group Address	Function
1/2/1	Line 1 ON/OFF
1/2/2	Line 2 ON/OFF
1/2/3	Line 3 ON/OFF
1/2/10	Block Line 1
1/2/11	Block Line 2
1/2/12	Block Line 3

5.2 HVAC

Group Address	Function
1/3/1	Heat / Cool Mode
1/3/2	Actual Temperature
1/3/3	Setpoint Display
1/3/4	Setpoint Value
1/3/10	HVAC Enable

6 Overall System Architecture

The system is based on a clear separation between:

- Decision layer (KNX)
- Execution layer (DALI / BACnet)
- Management layer (IP Application)

Interconnection between layers is achieved using gateways without transferring control logic outside KNX.

7 Message Exchange and Protocol Translation

This stage represents the core section of the project, where the mechanism of command and data exchange between different protocols is explained. It illustrates how gateways translate message formats (Telegram / Frame / Service) from one protocol to another while preserving the functional meaning of the command.

7.1 Message Translation from KNX to DALI

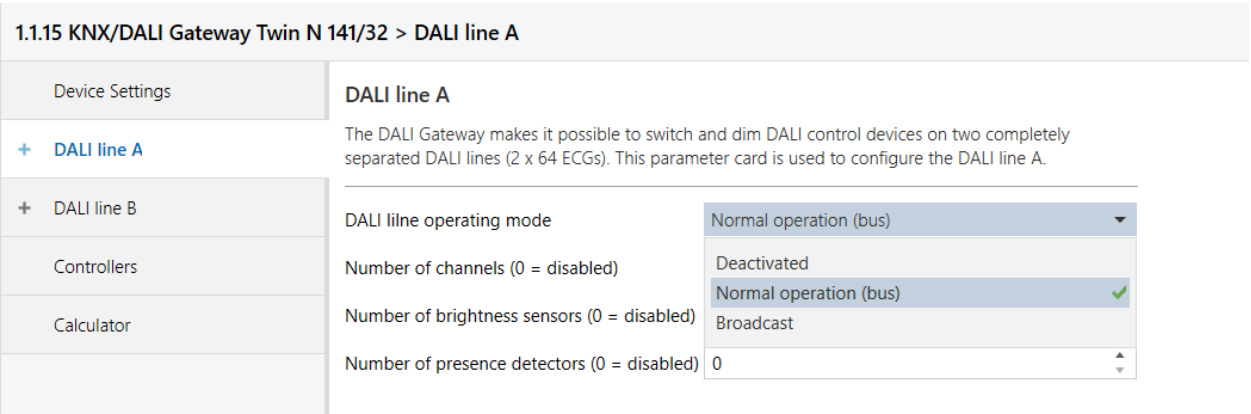
(KNX → DALI Message Translation)

Gateway name : 5WG1 141-1AB32

Steps :-

The KNX–DALI lighting gateway provides multiple configuration options and a wide range of functional objects that enable flexible and efficient lighting control. These objects include basic switching functions for ON/OFF control, precise dimming control for light intensity adjustment, as well as support for lighting scenarios and scenes. Through its configurable object structure, the gateway allows each lighting function to be mapped to corresponding KNX group addresses without embedding any control logic internally.

The following sequence illustrates how the gateway is configured to align with the requirements of the proposed system, ensuring that lighting commands originating from the KNX control layer are accurately translated and executed within the DALI lighting network while maintaining centralized decision-making within KNX :



1.1.15 KNX/DALI Gateway Twin N 141/32 > DALI line A

- Device Settings
- + DALI line A
- + DALI line B
- Controllers
- Calculator

DALI line A

The DALI Gateway makes it possible to switch and dim DALI control devices on two completely separated DALI lines (2 x 64 ECGs). This parameter card is used to configure the DALI line A.

DALI lilne operating mode	Normal operation (bus)
Number of channels (0 = disabled)	3
Number of scenes (0 = disabled)	0
Number of standby areas (0 = disabled)	0
Number of brightness sensors (0 = disabled)	0
Number of presence detectors (0 = disabled)	0

Number	Name	Object Function	Description	Group Addres	Length	C	R	W	T
1	Status device function	Ok/Defect			1 bit	C	R	-	T
2	Send status values	request			1 bit	C	-	W	-
11	A1 Switching	On/Off			1 bit	C	-	W	-
13	A1 Dimming	brighter/darker			4 bit	C	-	W	-
14	A1 Dimming value 1	8-bit value			1 byte	C	-	W	-
18	A1 Status switching	On/Off			1 bit	C	R	-	T
19	A1 Status dimming value	8-bit value			1 byte	C	R	-	T
138	A2 Switching	On/Off			1 bit	C	-	W	-
140	A2 Dimming	brighter/darker			4 bit	C	-	W	-
141	A2 Dimming value 1	8-bit value			1 byte	C	-	W	-
145	A2 Status switching	On/Off			1 bit	C	R	-	T
146	A2 Status dimming value	8-bit value			1 byte	C	R	-	T
265	A3 Switching	On/Off			1 bit	C	-	W	-
267	A3 Dimming	brighter/darker			4 bit	C	-	W	-
268	A3 Dimming value 1	8-bit value			1 byte	C	-	W	-
272	A3 Status switching	On/Off			1 bit	C	R	-	T
273	A3 Status dimming value	8-bit value			1 byte	C	R	-	T
8739	B1 Switching	On/Off			1 bit	C	-	W	-
8741	B1 Dimming	brighter/darker			4 bit	C	-	W	-
8742	B1 Dimming value 1	8-bit value			1 byte	C	-	W	-
8746	B1 Status switching	On/Off			1 bit	C	R	-	T
8747	B1 Status dimming value	8-bit value			1 byte	C	R	-	T
8866	B2 Switching	On/Off			1 bit	C	-	W	-
8868	B2 Dimming	brighter/darker			4 bit	C	-	W	-
8869	B2 Dimming value 1	8-bit value			1 byte	C	-	W	-
8873	B2 Status switching	On/Off			1 bit	C	R	-	T
8874	B2 Status dimming value	8-bit value			1 byte	C	R	-	T
8993	B3 Switching	On/Off			1 bit	C	-	W	-

Number	Name	Object Function	Description	Group Addres	Length	C	R	W	T
1	Status device function	Ok/Defect			1 bit	C	R	-	T
2	Send status values	request			1 bit	C	-	W	-
11	A1 Switching	On/Off	control line 1 on off	1/2/1	1 bit	C	-	W	-
13	A1 Dimming	brighter/darker			4 bit	C	-	W	-
14	A1 Dimming value 1	8-bit value			1 byte	C	-	W	-
18	A1 Status switching	On/Off			1 bit	C	R	-	T
19	A1 Status dimming value	8-bit value			1 byte	C	R	-	T
138	A2 Switching	On/Off	control line 2 on...	1/2/2	1 bit	C	-	W	-
140	A2 Dimming	brighter/darker			4 bit	C	-	W	-
141	A2 Dimming value 1	8-bit value			1 byte	C	-	W	-
145	A2 Status switching	On/Off			1 bit	C	R	-	T
146	A2 Status dimming value	8-bit value			1 byte	C	R	-	T
265	A3 Switching	On/Off	control line 3 on...	1/2/3	1 bit	C	-	W	-
267	A3 Dimming	brighter/darker			4 bit	C	-	W	-
268	A3 Dimming value 1	8-bit value			1 byte	C	-	W	-
272	A3 Status switching	On/Off			1 bit	C	R	-	T

7.1.1 Message Source (KNX Side)

In this case, the command originates from one of the KNX devices, such as:

- KNX Touch Panel
- KNX Push Button
- Logic Function within KNX

Example:
The user wants to switch ON the first lighting line (Lighting Line 1).
A KNX message (KNX Telegram) is sent in the following form:

- Group Address (GA): 1/2/1
- Data Point Type (DPT): 1.001
- Value: 1

Explanation of each element:
Group Address (1/2/1):
Represents the function of switching ON/OFF Lighting Line 1.
DPT 1.001:
A 1-bit Boolean data type commonly used for ON/OFF commands.
Value = 1:
Indicates ON.
This message is transmitted over the KNX TP Bus and reaches all devices associated with this Group Address.

7.1.2 Message Reception inside the KNX–DALI Gateway

The KNX–DALI Gateway operates as:

- A KNX Application Device on the KNX side
- A DALI Master on the DALI side

Step 1: Reception
The gateway receives the KNX Telegram via the KNX Bus and checks:

- Group Address

- Data type
- Transmitted value

Step 2: Mapping Logic
Within the gateway configuration tool, a predefined mapping table exists, such as:

KNX Group Address	DALI Mapping
1/2/1	Group 1
Function	ON / OFF

The gateway does not execute control logic; it performs translation only.

7.1.3 DALI Command Generation

After determining the mapping, the gateway generates the appropriate DALI command:

- Target: DALI Group 1
- Command: ON

Execution:
The command is transmitted over the DALI Bus (2-wire, polarity-free).
Each driver that is a member of Group 1:

- Receives the command
- Executes it immediately
- Switches ON the connected lighting fixture

7.1.4 Functional Summary

KNX	Gateway	DALI
GA 1/2/1 = 1	Translation	Group 1 ON

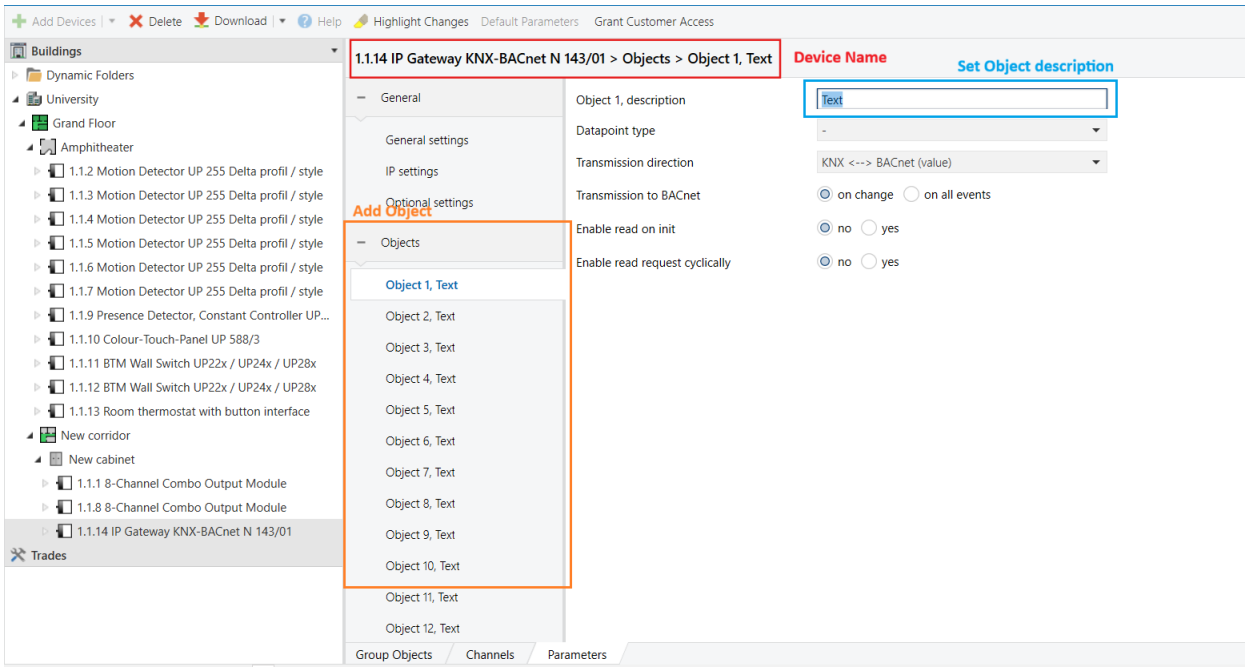
The functional meaning remains unchanged; only the message format and execution method are transformed.

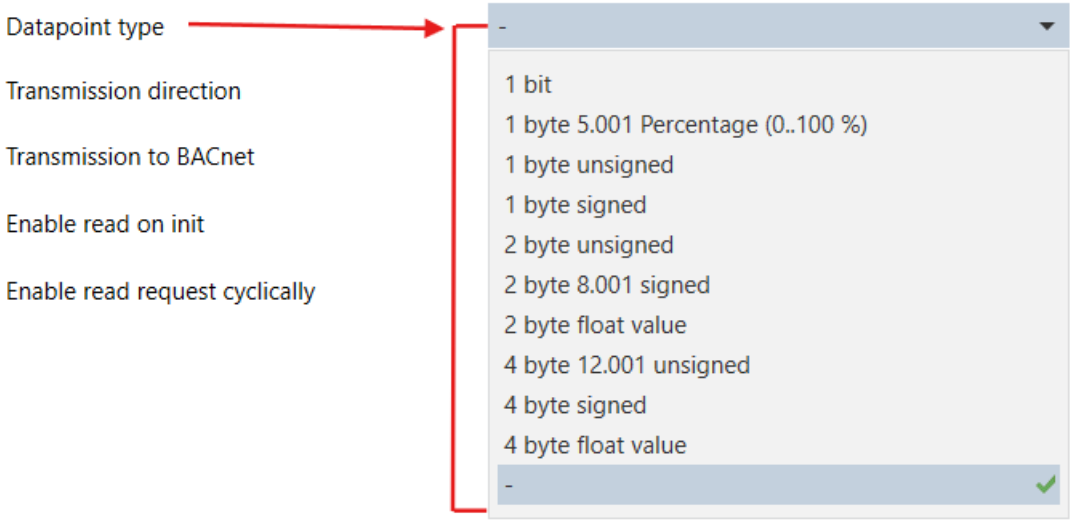
7.2 Message Translation from KNX to BACnet

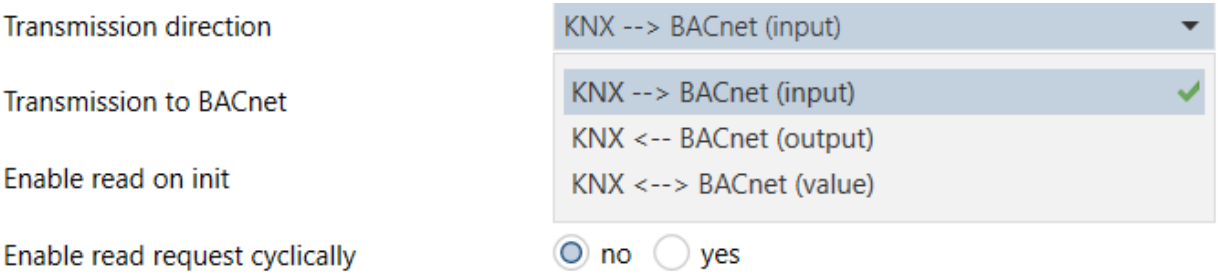
(KNX → BACnet Message Translation)

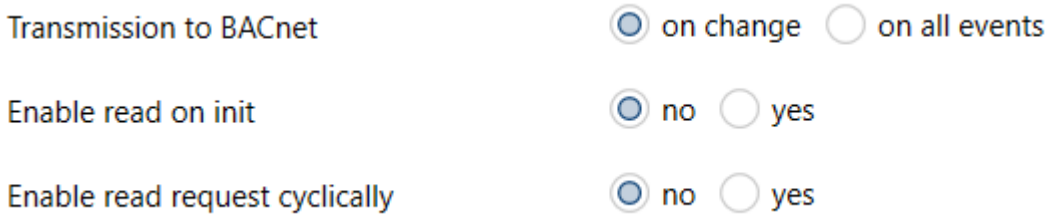
Gateway name : 5WG1_143_1AB01

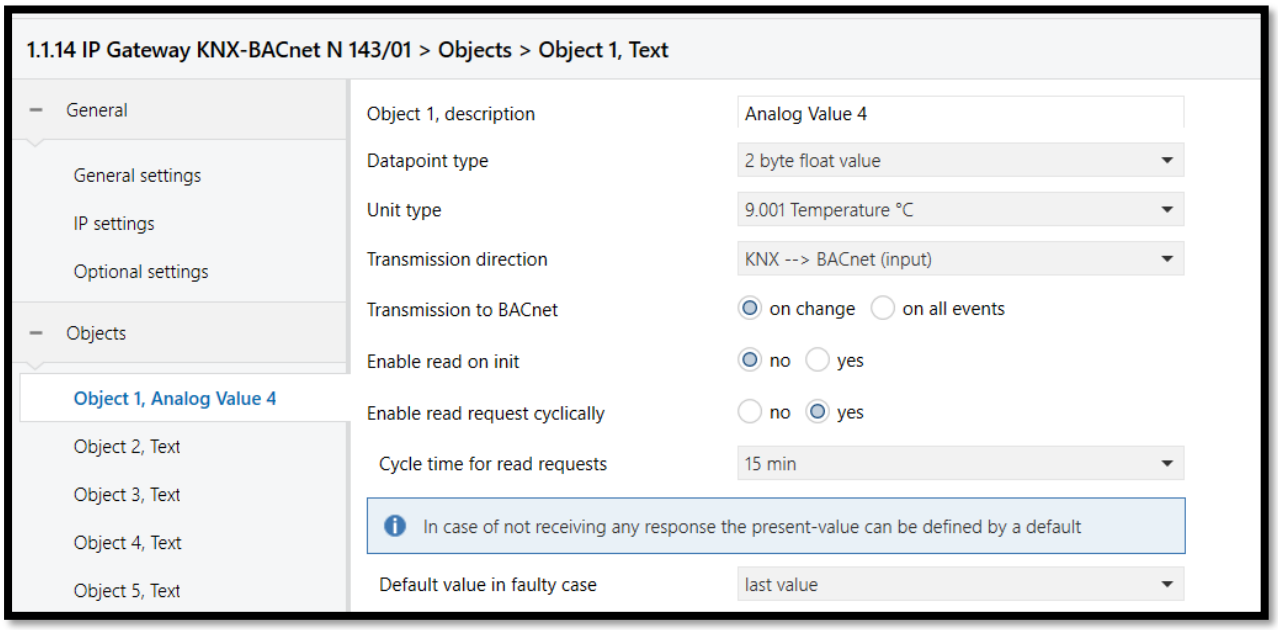
Steps :-




Datapoint type
Transmission direction
Transmission to BACnet
Enable read on init
Enable read request cyclically
-
1 bit
1 byte 5.001 Percentage (0..100 %)
1 byte unsigned
1 byte signed
2 byte unsigned
2 byte 8.001 signed
2 byte float value
4 byte 12.001 unsigned
4 byte signed
4 byte float value
-


Transmission direction
Transmission to BACnet
Enable read on init
Enable read request cyclically
KNX --> BACnet (input)
KNX --> BACnet (input)
KNX <-- BACnet (output)
KNX <--> BACnet (value)
no
yes


Transmission to BACnet
on change
on all events
Enable read on init
no
yes
Enable read request cyclically
no
yes


1.1.14 IP Gateway KNX-BACnet N 143/01 > Objects > Object 1, Text
General
General settings
IP settings
Optional settings
Objects
Object 1, Analog Value 4
Object 2, Text
Object 3, Text
Object 4, Text
Object 5, Text
Object 1, description
Analog Value 4
Datapoint type
2 byte float value
Unit type
9.001 Temperature °C
Transmission direction
KNX --> BACnet (input)
Transmission to BACnet
on change
on all events
Enable read on init
no
yes
Enable read request cyclically
no
yes
Cycle time for read requests
15 min
In case of not receiving any response the present-value can be defined by a default
Default value in faulty case
last value

7.2.1 Message Source (KNX Side)

In this scenario, the user wants to modify the HVAC system setpoint temperature.

KNX Telegram:

- Group Address: 1/3/4
- Value: 22.0

Explanation:

GA 1/3/4:

Represents the setpoint temperature.

Value = 22.0:

Desired temperature in degrees Celsius.

Data type:

Float / temperature-related DPT.

7.2.2 Message Reception inside the KNX–BACnet Gateway

The KNX–BACnet Gateway acts as a translation intermediary.

Step 1: Reception

The gateway receives the telegram from the KNX Bus and identifies the Group Address and its value.

Step 2: Object Mapping

KNX Group Address	BACnet Object
1/3/4	Analog Value (AV:4)

Any change on this Group Address is translated into a corresponding BACnet object update.

7.2.3 BACnet Service Generation

After mapping, the gateway sends the following BACnet message:

```
WriteProperty(
  Object = Analog Value:4,
  Property = Present_Value,
  Value = 22.0
)
```

Execution:

- The BACnet HVAC Controller receives the request
- The setpoint is updated internally
- The HVAC system operates according to the new value

7.2.4 Functional Summary

KNX	Gateway	BACnet
GA 1/3/4 = 22	Translation	AV:4.Present_Value = 22

7.3 Interaction Between IP Application and KNX Control Layer

(IP Application → KNX)
In this architecture, the IP Application does not communicate directly with execution protocols such as BACnet or DALI. Instead, it interacts exclusively with the KNX system, which acts as the central control and decision-making layer.
The IP Application functions as a visualization and supervisory software layer that allows the user to monitor system states and issue control commands by writing to KNX objects. All logical decisions and protocol translations are handled within the KNX ecosystem.

7.3.1 Role of the IP Application

The IP Application is responsible for:

- Data visualization (dashboards and graphical interfaces)
- User interaction and command input
- Monitoring system states

It does **not** contain control logic and does **not** communicate directly with BACnet or DALI devices.

7.3.2 Command Transmission from IP Application to KNX

When the user performs a control action (e.g., changing lighting state or HVAC setpoint), the IP Application sends a command to a KNX Group Address or KNX object via a KNX/IP Interface or KNX/IP Server.
Example: Lighting Control

```
Write Group Address
GA = 1/2/1
Value = 1
```

Explanation:

- The IP Application writes to a KNX Group Address
- The KNX system receives the telegram
- KNX logic evaluates the command

At this stage, the IP Application has no knowledge of the underlying execution protocol.

7.3.3 KNX as the Central Control Core

After receiving the command from the IP Application, KNX performs the following:

- Executes logical decisions (blocks, conditions, overrides)
- Determines the appropriate action
- Sends commands to the relevant gateway

Depending on the function:

- KNX–DALI Gateway translates the command to DALI for lighting control
- KNX–BACnet Gateway translates the command to BACnet for HVAC control

Thus, KNX remains the single point of control.

7.3.4 Monitoring and Feedback to IP Application

For monitoring purposes, the IP Application subscribes to KNX Group Addresses.
Example: Temperature Monitoring

```
GA = 1/3/2
Value = 24.5
```

- Sensors or gateways update KNX Group Addresses
- KNX/IP Interface forwards updates to the IP Application
- The IP Application displays the values to the user

No direct BACnet ReadProperty or SubscribeCOV services are used by the IP Application.

7.3.5 Why This Architecture Is Correct

Design Principle	Justification
KNX as Master Controller	Prevents command conflicts
IP as Visualization Layer	Improves stability and security
Gateways Perform Translation Only	Separation of logic and execution
No Direct IP–BACnet/DALI Control	Higher reliability

7.3.6 Functional Summary

Layer	Role	Communication
IP Application	Visualization and User Interface	KNX only
KNX	Control and Decision Logic	Gateways
Gateway	Protocol Translation	DALI / BACnet
DALI	Lighting Execution	Field Level
BACnet	HVAC Execution	Field Level

The IP Application does not control the building directly;
it controls KNX objects.
KNX controls the building.
Gateways translate commands without executing logic.

8 Conclusion

This project presented a comprehensive academic model for designing an integrated smart control system for a large university auditorium based on the integration of multiple standard protocols, while maintaining a clear separation between control logic, execution, and supervision layers.
The proposed design offers flexibility, scalability, and high reliability, making it suitable for both practical deployment and academic study.